Lesson 11.1 - Hypothesis Testing - Difference of Two Means

> **Video:** Hypothesis Testing - Difference of Two Means - Student's t-Distribution & Normal Distribution by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

The big picture: from one sample to two

In Chapter 10 we compared **one** sample with a number that H_0 claimed (for example $\mu = 100$ g). In this chapter there is no outside number. We compare **two groups with each other**: two teachers, two factories, two batches, before vs after.

1. The question: is the gap real, or just luck? Even if two teachers were exactly equally good, their class averages would still differ a little, because each class is a random sample of students. The test asks whether the gap we see is bigger than chance alone would usually produce.

2. H_0 says "no difference". H_0 : $\mu_1 - \mu_2 = 0$. This gives H_0 the one exact number it needs (Lesson 10.1): the difference is **0**. H_a says the difference is not 0 (or is < 0, or > 0).

3. Turn two samples into one number: the difference. Compute $\bar{x}_1 - \bar{x}_2$, then use the same recipe as Lesson 10.4:

$$\text{test statistic} = \frac{\text{what we got} - \text{what } H_0 \text{ claims}}{\text{standard error}} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE_{\text{difference}}}$$

4. Why the standard error gets bigger. Each sample mean has its own chance error. The difference picks up **both** of them, so it wobbles more than either mean alone. Variances add (never subtract):

$$SE_{\text{difference}} = \sqrt{SE_1^2 + SE_2^2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

In Example 1 below, $SE_1 = 0.62$ and $SE_2 = 0.49$, so $SE_{\text{difference}} = \sqrt{0.62^2 + 0.49^2} = 0.79$. That is bigger than either one, but smaller than simply adding them (1.11).

Where the formula comes from. It follows from three rules about variance, where $\operatorname{Var}(X) = E[(X - \mu)^2]$ is the square of the SD.

Rule 1: scaling a variable scales its variance by the square.

$$\operatorname{Var}(cX) = E[(cX - c\mu)^2] = c^2\, E[(X - \mu)^2] = c^2 \operatorname{Var}(X)$$

So $\operatorname{Var}(-Y) = (-1)^2 \operatorname{Var}(Y) = \operatorname{Var}(Y)$. Flipping the sign doesn't shrink the spread.

Rule 2: for independent variables, variances add. Expand the square:

$$\begin{aligned} \operatorname{Var}(X + Y) &= E\big[\big((X - \mu_X) + (Y - \mu_Y)\big)^2\big] \\ &= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\underbrace{E[(X - \mu_X)(Y - \mu_Y)]}_{\operatorname{Cov}(X,\,Y)} \end{aligned}$$

If X and Y are independent, $\operatorname{Cov}(X, Y) = 0$, so $\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$.

Rule 3: so does the variance of a difference. Write $X - Y = X + (-Y)$ and use Rules 1 and 2:

$$\operatorname{Var}(X - Y) = \operatorname{Var}(X) + \operatorname{Var}(-Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$$

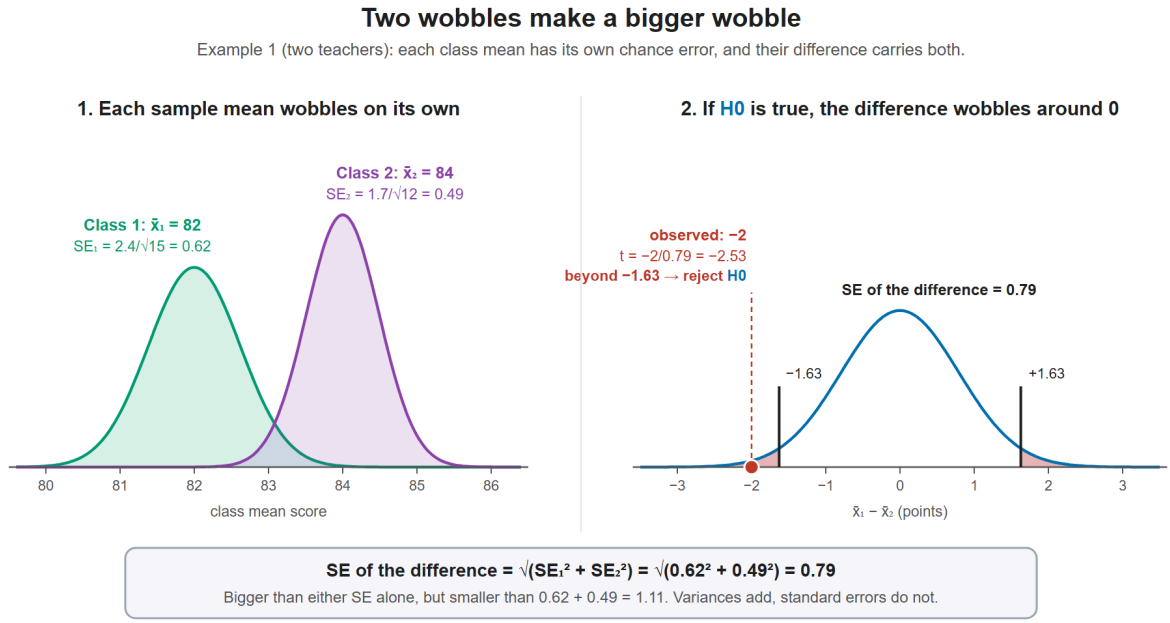
Now apply the rules to sample means.

- **One mean.** $\bar{x} = (X_1 + \cdots + X_n)/n$, with independent X_i that each have variance σ^2. Rule 2 gives $\mathrm{Var}(X_1 + \cdots + X_n) = n\sigma^2$. Rule 1 with $c = 1/n$ then gives $\mathrm{Var}(\bar{x}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$, so $SE = \sigma/\sqrt{n}$.
- **Two means.** The samples are independent, so Rule 3 gives:

$$\mathrm{Var}(\bar{x}_1 - \bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \quad \Rightarrow \quad SE_{\text{difference}} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{SE_1^2 + SE_2^2}$$

We don't know σ_1 and σ_2, so we plug in s_1 and s_2. That's why the test uses t.

- **Like a right triangle.** Variances (squares) add, not SDs, so the SEs combine like the two shorter sides of a right triangle: $\sqrt{0.62^2 + 0.49^2} = 0.79$ is the long side. That's why it is bigger than each SE but less than their sum.
- **When the groups are not independent** (the same people measured twice), the covariance term stays: $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y)$. Before and after weights move together, so the covariance is large and positive, and subtracting it removes most of the variance. That's why pairing helps (Lesson 11.4).



Left: each class mean has its own wobble. Right: if H_0 is true, the difference $\bar{x}_1 - \bar{x}_2$ wobbles around 0, and the wobble is wider than either one alone. The observed difference, -2, is 2.53 standard errors below 0, beyond the critical value, so we reject H_0.

5. "Independent" matters. This lesson assumes two separate groups of different people, with nothing linking someone in group 1 to someone in group 2. If the **same** people are measured twice (before and after), the groups are paired. Use Lesson 11.4 instead.

Which lesson for which question (Chapter 11 map):

What are you comparing?	Lesson	Tool
Means of two **independent** groups	11.1 (this one)	t (Welch) or z
How big is the difference between two means?	11.2	Cohen's d
Proportions (yes/no data) of two groups	11.3	z with a pooled $\bar{p}$
The **same subjects twice**, or matched pairs	11.4	paired t on the differences

The 5 steps are the same as in Chapter 10. Only the formula in step 4 changes:

1. Write H_0 (difference $= 0$) and H_a.
2. Choose α.
3. Use the sign in H_a to find the critical value(s).
4. **Compute the test statistic from both samples.** ← this lesson
5. Compare: beyond the critical value → reject H_0. Otherwise → fail to reject H_0.

Setup

Two **independent** groups. Does their population mean differ?

$$H_0 : \mu_1 = \mu_2 \;\; (\mu_1 - \mu_2 = 0) \qquad H_a : \mu_1 \neq \mu_2 \;\; (\mu_1 - \mu_2 \neq 0)$$

Test statistic

$$t \text{ (or } z) = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$$

Under H_0, $\mu_1 - \mu_2 = 0$. The denominator is the **standard error of the difference**.

Sample sizes	Use
Small (< 30)	**t-distribution** (Welch), with the degrees of freedom below
Large (≥ 30)	**z** (normal distribution)

Degrees of freedom (Welch-Satterthwaite):

$$\text{df} = \frac{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)^2}{\dfrac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \dfrac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

Example 1 - two teachers (t-test)

Do two teachers' classes score differently? $\alpha = 0.05$.

	Class 1	Class 2
n	15	12
$\bar{x}$	82	84
s	2.4	1.7

Hypotheses: $H_0 : \mu_1 - \mu_2 = 0$, $H_a : \mu_1 - \mu_2 \neq 0$ (two-tailed, $\alpha/2 = 0.025$).

Both samples are small → t-test.

Degrees of freedom:

- $s_1^2/n_1 = 5.76/15 = 0.384$ and $s_2^2/n_2 = 2.89/12 = 0.24083$

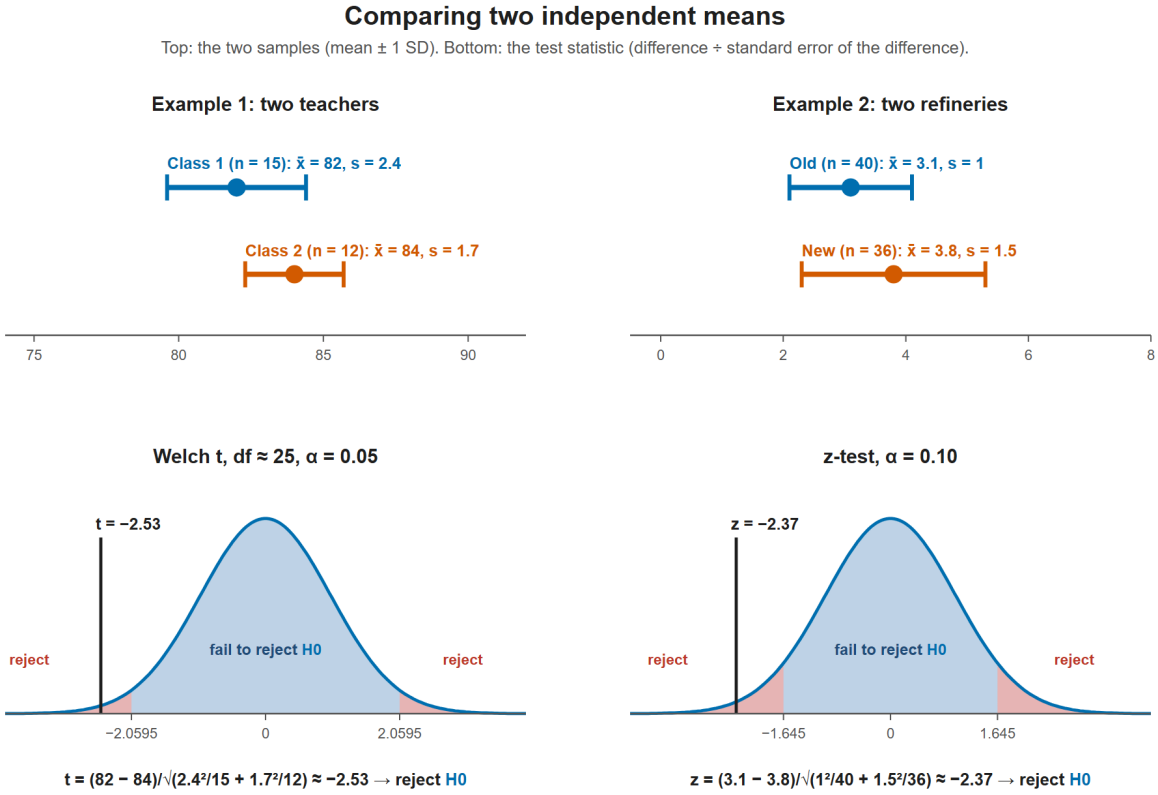
- Numerator: $(0.384 + 0.24083)^2 = (0.62483)^2 = 0.390417$
- Denominator: $\frac{0.384^2}{14} + \frac{0.24083^2}{11} = 0.010533 + 0.005273 = 0.015805$
- $\text{df} = \frac{\text{numerator}}{\text{denominator}} = 0.390417/0.015805 \approx 24.7$

The video rounds this to **25**, giving $t_{0.025,\,25} = 2.0595$. Many textbooks round df **down** to be conservative ($24 \rightarrow t = 2.064$). Both give the same decision here.

Test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{82 - 84 - 0}{\sqrt{0.384 + 0.24083}} = \frac{-2}{0.79046} \approx -2.53$$

$-2.53 < -2.0595$, which is in the rejection region. **Reject H_0.** At the 5% level there is a significant difference between the two teachers' class means.



Top: the two samples in each example. Bottom: the difference in means divided by its standard error. In both examples the statistic lands in the left rejection region, so the means differ significantly.

Example 2 - two oil refineries (z-test)

Is there a difference in refining rate (liters per second)? $\alpha = 0.10$.

	Old factory	New factory
n	40	36
$\bar{x}$	3.1	3.8
s	1.0	1.5

Hypotheses: H_0 : $\mu_1 = \mu_2$, H_a : $\mu_1 \neq \mu_2$ (two-tailed).

Both samples are large → z-test. The confidence level is 90%, so the left area for the critical value is 0.95: $z = \pm 1.645$.

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{3.1 - 3.8}{\sqrt{\frac{1^2}{40} + \frac{1.5^2}{36}}} = \frac{-0.7}{\sqrt{0.025 + 0.0625}} = \frac{-0.7}{0.2958} \approx -2.37$$

$-2.37 < -1.645$ → **reject** H_0. At the 10% level there is a significant difference between the factories; the new one refines faster.

Key takeaways

- H_0: equal means (difference 0). Test statistic = (difference in sample means) ÷ $\sqrt{s_1^2/n_1 + s_2^2/n_2}$.
- The standard error of a difference is $\sqrt{SE_1^2 + SE_2^2}$: two sources of chance error, so it is bigger than either one alone.
- Small samples → t with Welch's df. Large samples → z.
- Compare with the critical value(s), or with the p-value, as in the one-sample case.

Lesson 11.2 - How to Find Cohen's D to Determine the Effect Size

> **Video:** How to find Cohen's D to determine the Effect Size Using Pooled Standard Deviation by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

The big picture: is the difference big, not just real?

Lesson 11.1 asked **whether** two means differ. This lesson asks **how much** they differ, and why that is a separate question.

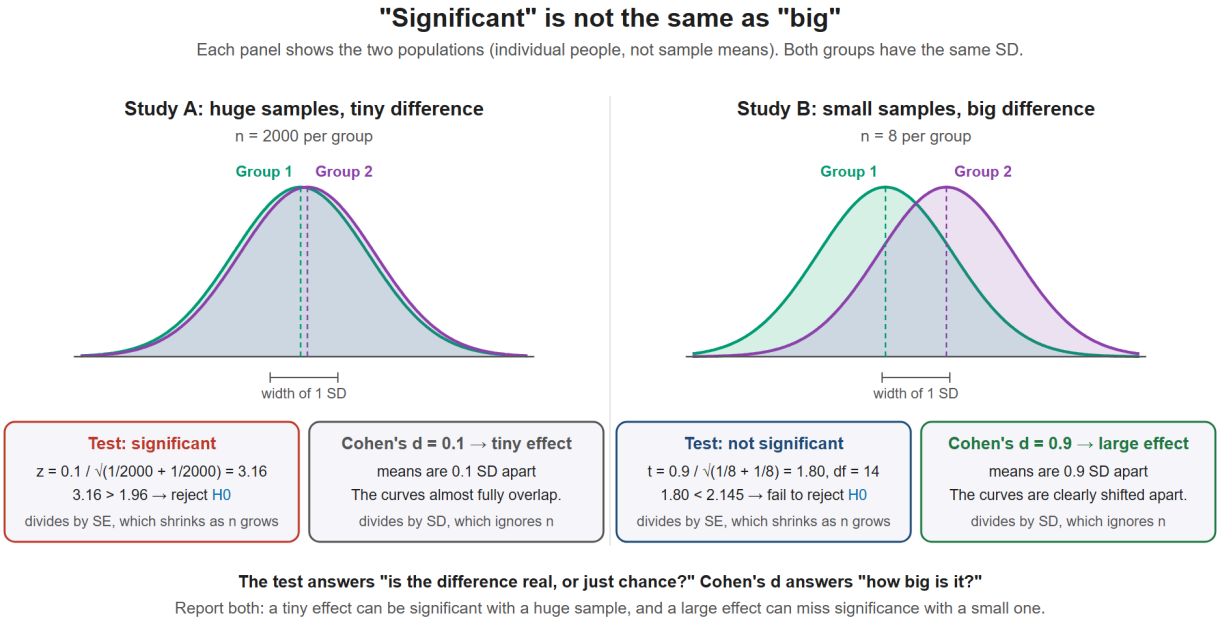
1. A test answers "real or luck?", not "big or small?". Rejecting H_0 means the gap is too large to blame on chance. It doesn't say the gap is large enough to matter.

2. Significance depends on sample size. Lesson 10.4 showed that the same gap gives a bigger test statistic with a bigger sample, because the standard error shrinks like $1/\sqrt{n}$. So:

- With **huge** samples, even a tiny, unimportant difference becomes "significant".
- With **small** samples, even a large, important difference can fail to reach significance.

3. Cohen's d removes the sample size. It divides by the **standard deviation** (how much individual people vary), not the **standard error** (how much sample means vary):

	Test statistic (Lesson 11.1)	Cohen's d (this lesson)
Divides the gap by	standard **error**, $\approx s\sqrt{1/n_1 + 1/n_2}$	standard **deviation**, s_{pooled}
Gets bigger with more data?	**Yes**, because the SE shrinks	**No**, because the SD stays about the same
Answers	Is there a difference?	How big is the difference?



Study A has a tiny effect ($d = 0.1$), but with 2000 people per group it is still significant. Study B has a large effect ($d = 0.9$), but with only 8 people per group it misses significance. The test and Cohen's d answer different questions.

4. Why "standard deviation units"? Raw gaps depend on the units: 3 points on a test, 4 pounds of weight. Dividing by the SD makes d **unitless**, so you can compare effects across different studies and measurements. $d = 0.5$ means the same thing everywhere: the averages are half a standard deviation apart.

5. Why pool the standard deviations? d needs **one** SD to divide by, but each group has its own s. Assuming both groups vary about the same amount, we combine them into a single best estimate, giving the bigger group more weight.

> **Report both.** A full answer says whether the difference is real (test or p-value) **and** how big it is (Cohen's d). For example: "significant at the 5% level, with a large effect, $d = 0.85$."

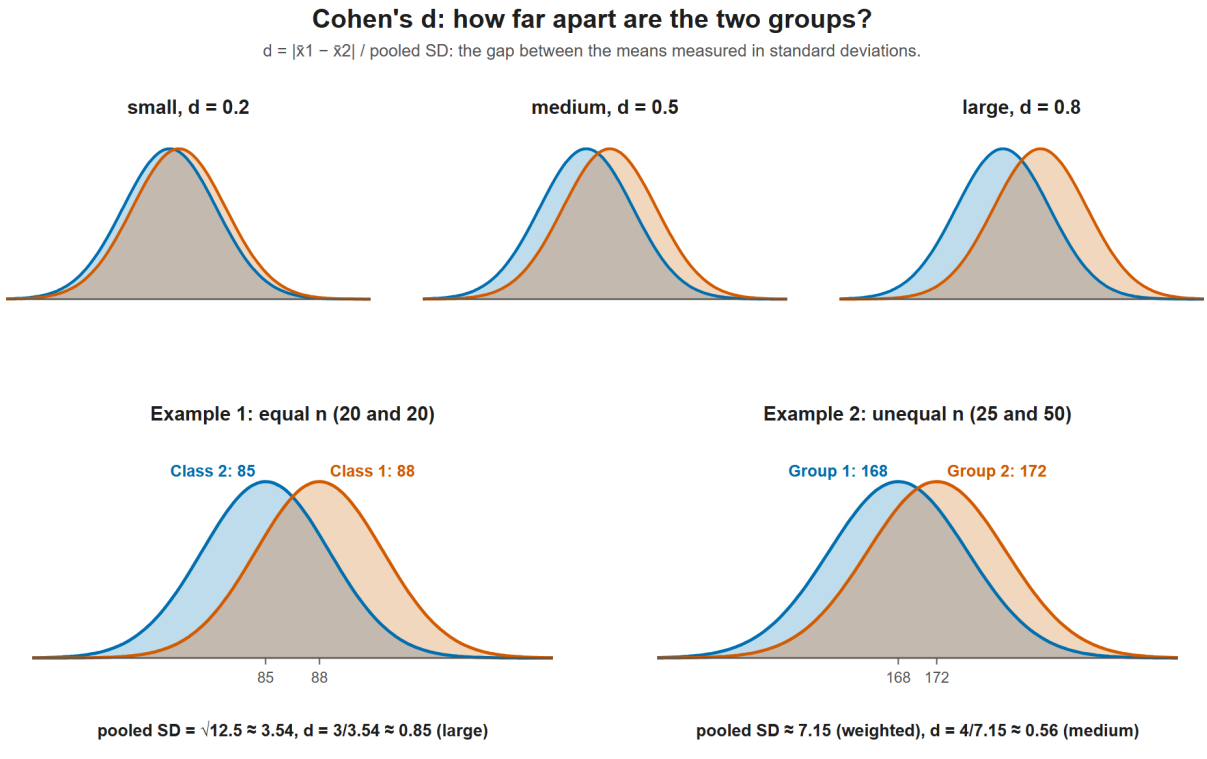
What Cohen's d measures

Cohen's d expresses the difference between two means **in standard-deviation units**:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}}$$

- $d = 1$: the means are 1 standard deviation apart.
- $d = 2$: 2 standard deviations apart.
- $d = 0.5$: half a standard deviation apart.

d	Effect size
≈ 0.2	Small
≈ 0.5	Medium
≥ 0.8	Large



Cohen's d measures how far apart the two curves sit, in units of the pooled standard deviation. The more the curves overlap, the smaller the effect. Example 1 (d ≈ 0.85) is a large effect; Example 2 (d ≈ 0.56) is medium.

Pooled standard deviation

General formula (use it when the sample sizes differ):

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Shortcut (only when $n_1 = n_2$, or nearly so):

$$s_{\text{pooled}} = \sqrt{\frac{s_1^2 + s_2^2}{2}}$$

Example 1 - two classes, equal sample sizes

	Class 1	Class 2
n	20	20
$\bar{x}$	88	85
s	3	4

Pooled SD (general formula):

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{19(3^2) + 19(4^2)}{20 + 20 - 2}} = \sqrt{\frac{171 + 304}{38}} = \sqrt{\frac{475}{38}} = \sqrt{12.5} \approx 3.5355$$

Shortcut (equal n): $\sqrt{\frac{s_1^2 + s_2^2}{2}} = \sqrt{\frac{9 + 16}{2}} = \sqrt{12.5} \approx 3.5355$, the same answer ✓

Cohen's d:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}} = \frac{88 - 85}{3.5355} \approx 0.85 \;\Rightarrow\; \textbf{large effect}$$

Example 2 - two groups, unequal sample sizes

	Group 1	Group 2
n	25	50
$\bar{x}$ (lb)	168	172
s (lb)	5	8

Pooled SD (must use the general formula, because the sizes differ):

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{24(5^2) + 49(8^2)}{25 + 50 - 2}} = \sqrt{\frac{600 + 3136}{73}} = \sqrt{\frac{3736}{73}} \approx 7.154$$

The shortcut would give $\sqrt{(25 + 64)/2} \approx 6.671$, which is **wrong** here. Group 2 is twice as large, so its SD (8) should carry more weight, and the correct pooled SD (7.15) is closer to 8.

Cohen's d:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}} = \frac{|168 - 172|}{7.154} = \frac{4}{7.154} \approx 0.56 \;\Rightarrow\; \textbf{medium effect}$$

Key takeaways

- d = |difference in means| ÷ pooled SD, a **unitless** measure of how big the difference is.
- Benchmarks: 0.2 small, 0.5 medium, 0.8+ large.
- Use the weighted pooled-SD formula unless the sample sizes are equal.
- A hypothesis test asks *whether* there is a difference. Cohen's d asks *how large* it is.
- The test divides by the SE, which shrinks as n grows. d divides by the SD, which doesn't. So "significant" doesn't mean "big", and "not significant" doesn't mean "small".

Lesson 11.3 - Hypothesis Testing with Two Proportions

> **Video:** Hypothesis Testing With Two Proportions by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

The big picture: comparing two rates

Lessons 11.1 and 11.2 compared **averages** (scores, weights). Many questions are about **yes/no** data instead: defective or not, passed or failed, clicked or didn't. Each group is then summarized by a **proportion** (a rate), and we ask whether the two rates really differ.

1. The question. Batch 1 has 4% defective laptops and batch 2 has 6%. Is batch 2 really worse, or could a 2-point gap happen by chance with samples this size?

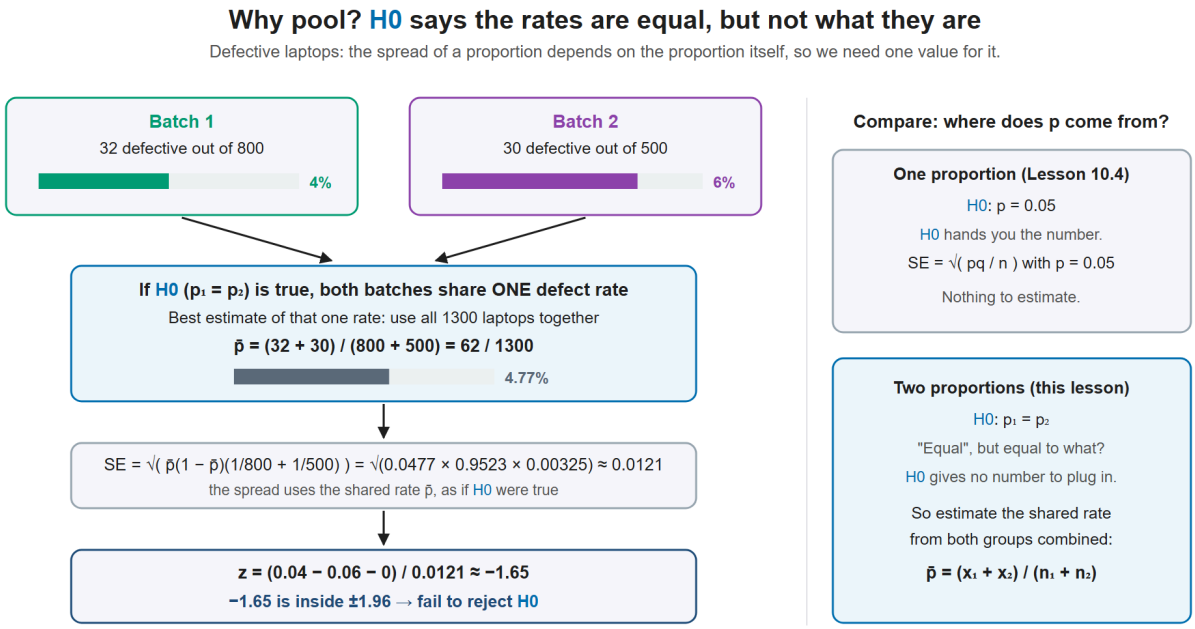
2. H_0 says "no difference". $H_0 : p_1 = p_2$, which is the same as $p_1 - p_2 = 0$. The recipe is the same as always:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{SE}$$

3. Why pool? H_0 says "equal", but not equal to what. For proportions, the spread depends on the proportion itself: $SE = \sqrt{p(1-p)/n}$. A rate near 50% wobbles more than a rate near 4%. So we need a value of p to compute the SE.

- **One proportion (Lesson 10.4):** $H_0 : p = 0.05$ hands us the number. Plug it in.
- **Two proportions (this lesson):** $H_0 : p_1 = p_2$ gives no number. But if H_0 is true, both groups share **one** rate, and the best estimate of that rate uses **all** the data together:

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{\text{total successes}}{\text{total sample size}}$$



If H_0 is true, the two batches are really one population, so we estimate its single defect rate from all 1300 laptops. That pooled rate goes into the standard error.

4. Why no pooling for means (Lesson 11.1)? A mean and its spread are separate numbers. $H_0 : \mu_1 = \mu_2$ says nothing about s_1 and s_2, so each group keeps its own. For proportions, the spread is tied to p, so assuming

equal p's also means assuming equal spreads.

5. Why z, not t? There is no separate s to estimate, because the spread comes straight from $\bar{p}$. With large samples the sample proportions are approximately normal, so we use z. A common check is at least about 10 successes and 10 failures in each group. Here there are 32 and 30 defectives, so that's fine.

Formulas

Quantity	Formula
Sample proportions	$\hat{p}_1 = \frac{x_1}{n_1},\ \hat{p}_2 = \frac{x_2}{n_2}$
Pooled proportion	$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$
Test statistic	$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\,(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

Under $H_0 : p_1 = p_2$, the term $p_1 - p_2$ is **0**. The proportions are pooled because H_0 assumes both groups share one common proportion. Large samples → use the normal (z) distribution.

Example - defective laptops

Two batches of laptops were tested: 32 of 800 were defective in group 1 and 30 of 500 in group 2. Is the difference significant at $\alpha = 0.05$?

Hypotheses (the test is run because someone suspects a difference):

$$H_0 : p_1 = p_2 \qquad H_a : p_1 \neq p_2$$

Sample proportions:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{32}{800} = 0.04, \qquad \hat{p}_2 = \frac{x_2}{n_2} = \frac{30}{500} = 0.06$$

Critical values (two-tailed, $\alpha/2 = 0.025$ per tail, left area 0.975): $z = \pm 1.96$

Pooled proportion:

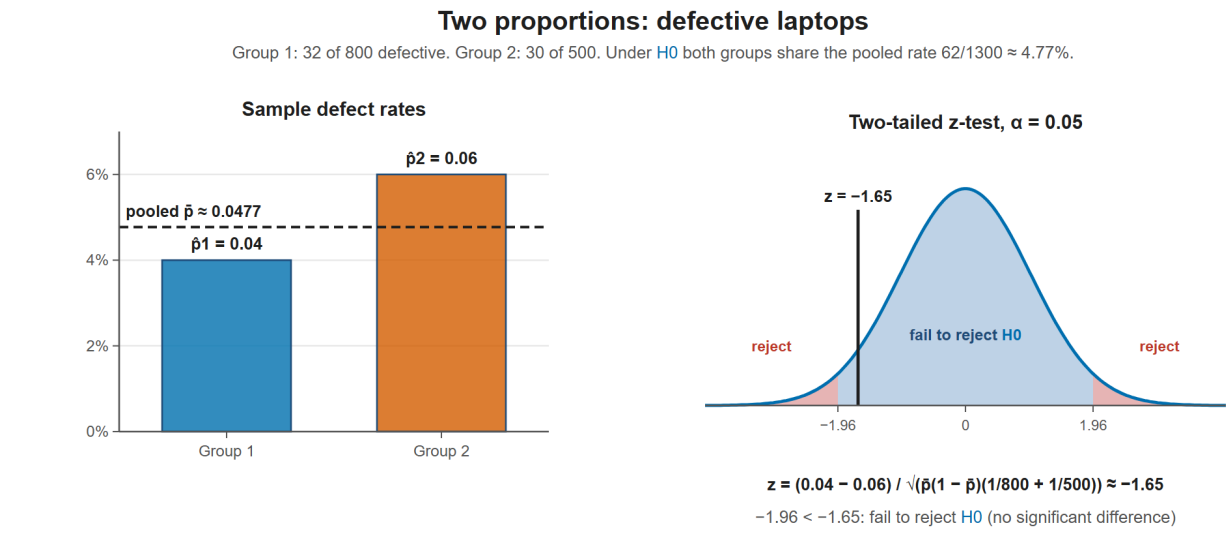
$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{32 + 30}{800 + 500} = \frac{62}{1300} \approx 0.04769$$

Test statistic:

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.04 - 0.06 - 0}{\sqrt{0.04769(1 - 0.04769)\left(\frac{1}{800} + \frac{1}{500}\right)}} = \frac{-0.02}{0.012149} \approx -1.65$$

$-1.96 < -1.65 < 0$, which is in the fail-to-reject region.

Fail to reject H_0. At the 5% level the difference in defect rates (4% vs 6%) is **not statistically significant**. This doesn't prove the rates are equal; the data just don't show a clear difference.



The two sample rates differ by 2 percentage points, but with these sample sizes that gives only z ≈ −1.65, which is inside ±1.96. The difference is not statistically significant.

Key takeaways

- We pool because H_0 says the rates are equal without saying what they equal, and the SE of a proportion needs a value of p.
- H_0 : $p_1 = p_2$. Use the **pooled** proportion $\bar{p} = (x_1 + x_2)/(n_1 + n_2)$ in the standard error.
- $z = (\hat{p}_1 - \hat{p}_2)/\sqrt{\bar{p}(1-\bar{p})(1/n_1 + 1/n_2)}$.
- Compare with the critical value(s) or use a p-value. Fail to reject means "no significant difference found", not "proven equal".

Lesson 11.4 - Matched or Paired Samples

> **Video:** Matched or Paired Samples T-Test - Hypothesis Testing by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

The big picture: why pairing helps

Lesson 11.1 compared two **separate** groups of people. Here the two columns of numbers come from the **same** people, weighed before and after a program. This is a different kind of data and needs a different test.

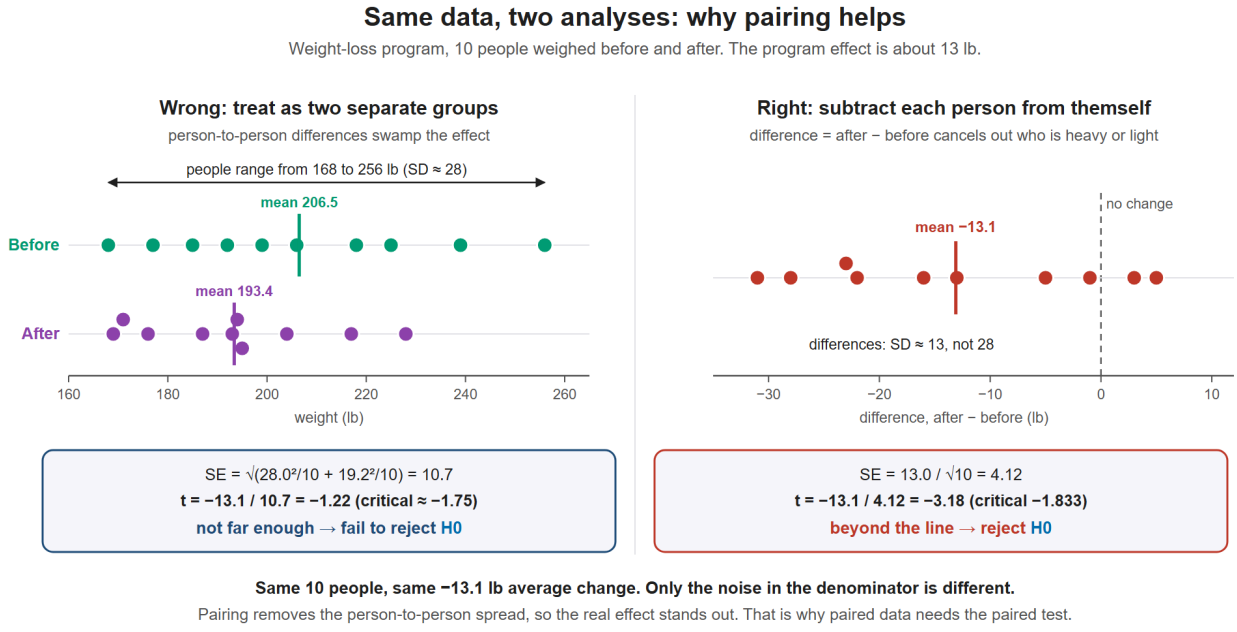
1. The two columns are linked. A heavy person before the program is still heavy after it. Row 7 (256 → 228) and row 6 (168 → 171) are different people, but each row is the same person measured twice. The data come in **pairs**.

2. People differ much more than the program does. Before the program, weights range from 168 to 256 lb, with an SD of about 28 lb. The program changes weight by only about 13 lb. If we treat before and after as two separate groups (Lesson 11.1), that 28 lb person-to-person spread goes into the standard error and hides the effect:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{28.0^2}{10} + \frac{19.2^2}{10}} = 10.7, \qquad t = \frac{\bar{x}_{\text{after}} - \bar{x}_{\text{before}}}{SE} = \frac{-13.1}{10.7} = -1.22 \;\Rightarrow\; \text{fail to reject } H_0$$

3. Subtract each person from themself. Taking $x_d = \text{after} - \text{before}$ cancels out who is heavy and who is light, leaving only each person's **change**. The differences vary much less (SD ≈ 13 lb):

$$SE = \frac{s_d}{\sqrt{n}} = \frac{13.0}{\sqrt{10}} = 4.12, \qquad t = \frac{\bar{x}_d}{SE} = \frac{-13.1}{4.12} = -3.18 \;\Rightarrow\; \text{reject } H_0$$



Same 10 people and the same average change of −13.1 lb. Only the noise in the denominator is different. Treating the data as two separate groups lets person-to-person differences drown out the effect. Pairing removes them.

4. A paired test is just a one-sample test on the differences. Once you have the column of differences, the original pairs are no longer needed. Run the Lesson 10.4 t-test on x_d:

- H_0: $\mu_d = 0$ (no change on average), or $\mu_d \geq 0$ for a one-sided claim.
- n = the number of **pairs**, so df $= n - 1$.

5. How to spot paired data. Ask: *does each value in group 1 have a specific partner in group 2?*

Paired (this lesson)	Independent (Lesson 11.1)
Same people before and after	Two different groups of people
Same subject under two conditions (left hand vs right hand)	Treatment group vs a separate control group
Twins, or deliberately matched pairs	Two classes, two factories, two batches

> **Using the wrong test matters.** Running an independent test on paired data ignores the pairing and usually misses real effects, as in step 2. Running a paired test on independent data is meaningless, because there are no real partners to subtract.

When samples are paired

A **matched (paired) sample** measures the **same subjects twice**, for example before and after a treatment. Instead of comparing two separate groups, analyze the **difference** for each subject:

$$x_d = \text{after} - \text{before}$$

Formulas

Quantity	Formula
Mean difference	$\bar{x}_d = \dfrac{\sum x_d}{n}$
SD of the differences	$s_d = \sqrt{\dfrac{\sum (x_d - \bar{x}_d)^2}{n-1}}$ (or `=STDEV.S(...)` in Excel)
Test statistic	$t = \dfrac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}}$, df $= n - 1$
Confidence interval	$\bar{x}_d \pm t\,\dfrac{s_d}{\sqrt{n}}$

Example - a weight-loss program

Is the program effective at reducing weight? $\alpha = 0.05$, 10 subjects.

Subject	Before (lb)	After (lb)	x_d = after − before
1	185	169	−16
2	192	187	−5
3	206	193	−13
4	177	176	−1
5	225	194	−31
6	168	171	+3
7	256	228	−28
8	239	217	−22
9	199	204	+5
10	218	195	−23

A negative difference means the subject lost weight.

Hypotheses. An effective program means the mean difference is **negative**:

$$H_0 : \mu_d \geq 0 \qquad H_a : \mu_d < 0 \quad \text{(left-tailed)}$$

Sample statistics:

$$\bar{x}_d = \frac{\sum x_d}{n} = \frac{-131}{10} = -13.1, \qquad s_d = \sqrt{\frac{\sum (x_d - \bar{x}_d)^2}{n-1}} = \sqrt{\frac{1526.9}{9}} \approx 13.025$$

Critical value: left-tailed, df $= 9$, $\alpha = 0.05$ gives $t = -1.833$.

Test statistic:

$$t = \frac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}} = \frac{-13.1 - 0}{13.025/\sqrt{10}} = \frac{-13.1}{4.119} \approx -3.18$$

$-3.18 < -1.833$, which is in the rejection region. **Reject H_0.** There is sufficient evidence at the 5% level that the program reduces weight.

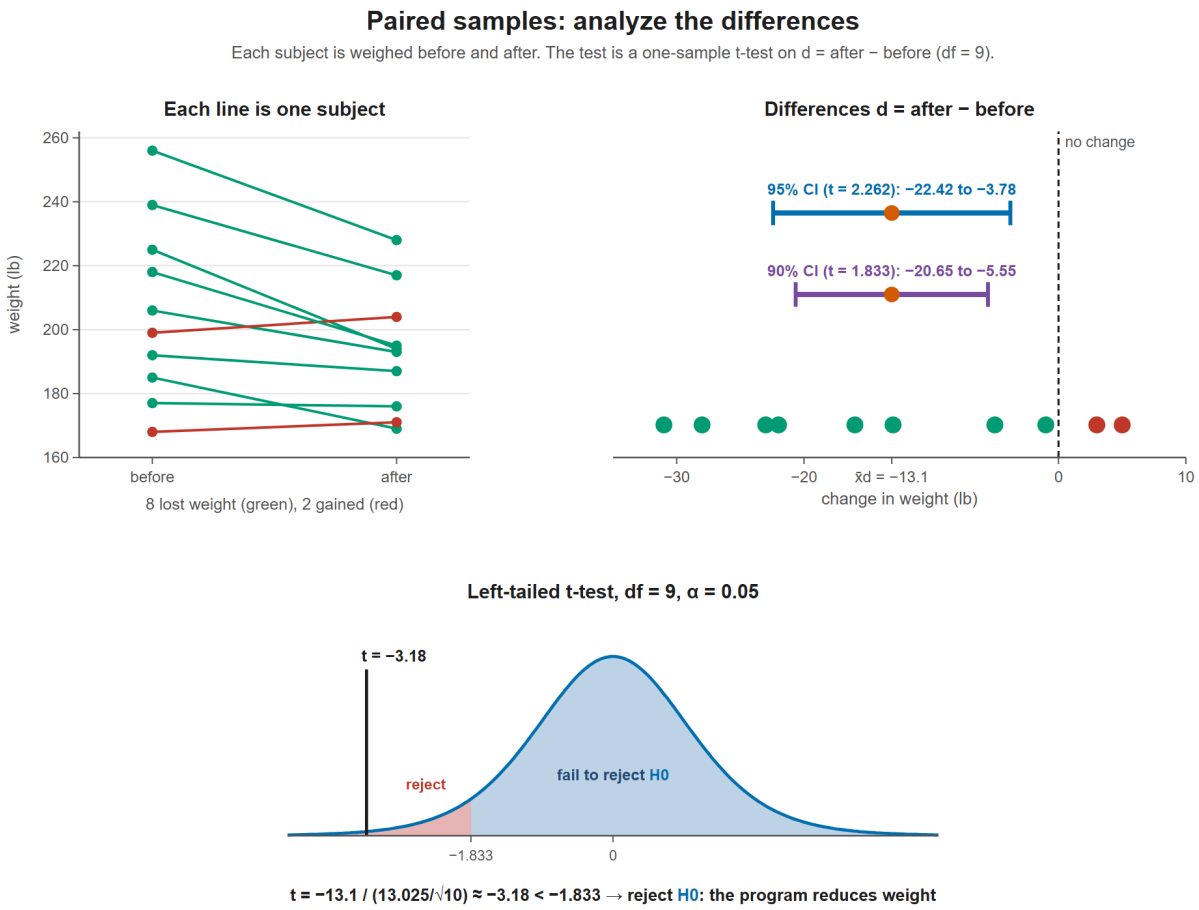
Confidence interval for μ_d

Using $t = 1.833$ (as in the video):

$$\bar{x}_d \pm t\,\frac{s_d}{\sqrt{n}} = -13.1 \pm 1.833\,(4.119) = -13.1 \pm 7.55 \;\Rightarrow\; (-20.65,\ -5.55)$$

> **Correction:** 1.833 is the critical value with **5% in one tail**, so a two-sided interval built from it is a **90%** confidence interval, not 95%. For a **95%** interval, use $t_{0.025,\,9} = 2.262$: $\bar{x}_d \pm t\,\frac{s_d}{\sqrt{n}} = -13.1 \pm 2.262(4.119) = -13.1 \pm 9.32 \Rightarrow (-22.42,\ -3.78)$.

Either way, the whole interval is **below 0**: there are no zero or positive values, which supports rejecting H_0. The margin of error is 7.55 for the 90% interval, or 9.32 for 95%.



Top left: most subjects lost weight. Top right: the ten differences, with the 90% and 95% intervals for the mean difference; both sit entirely below 0. Bottom: t = −3.18 is beyond the critical value −1.833, so H_0 is rejected.

Key takeaways

- Paired data → work with the **differences** x_d and run a one-sample t-test on them.
- Pairing removes person-to-person variation, so the SE is smaller and real effects are easier to detect (here $t = -3.18$ paired vs -1.22 if wrongly treated as independent).
- df $= n - 1$, where n is the number of **pairs**.
- Choose the direction of H_a to match the claim (e.g. "reduces" → $\mu_d < 0$ when x_d = after − before).
- Match the CI's t to its confidence level: two-sided 95% uses $\alpha/2 = 0.025$.